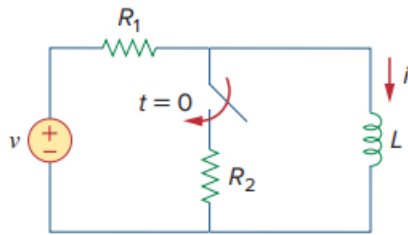


Problem

Using Fig. 7.118, design a problem to help other students better understand the step response of an RL circuit.

Fig. 7.118:



Step-by-step solution

Step 1 of 5

Refer to Figure 7.118 in the textbook.

The input voltage to the circuit is step function of magnitude V , as follows :

$$v = Vu(t) \text{ V}$$

The response of the circuit to a step input is the step response of the circuit.

The switch is closed at the time instant, $t = 0$.

The unit step response of the system is,

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-\frac{t}{\tau}}$$

Step 2 of 5

The initial value of the current through the inductor is, $i(0)$.

The steady value of the current through the inductor is, $i(\infty)$.

Since, the source is not connected to the circuit, the current through the inductor prior to the time instant $t = 0$ or $t < 0$, 0 A .

That is, $i(0^-) = 0 \text{ A}$.

The current through the inductor cannot change instantaneously; So the current through the inductor at $t = 0^-$ should be equal to current through it at 0^+ , that is, $i(0^-) = i(0^+)$.

Thus, the initial value of the inductor current is, $i(0) = 0 \text{ A}$; The inductor acts as short circuit at steady state, $t \rightarrow \infty$.

The resistor R_2 is irrelevant at steady state; since it is in parallel connection to a short circuit and the resultant resistance is zero.

The steady current through the inductor is,

$$i(\infty) = \frac{V}{R_1}$$

Step 3 of 5

The expression for time constant of a $R-L$ circuit is,

$$\tau = \frac{L}{R}$$

Determine the value of R for $t > 0$.

Short circuit the voltage source present in the circuit.

Remove the inductor from the circuit and calculate the equivalent resistance across the terminals of the inductor.

The equivalent resistance seen by the inductor is the parallel combination of R_1 and R_2 .

The parallel combination of R_1 and R_2 is,

$$R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

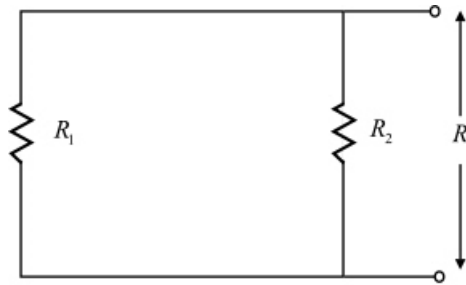


Figure 1: Equivalent resistance across the inductor.

Step 4 of 5

Determine the time constant of the circuit.

Substitute $\frac{R_1 \cdot R_2}{R_1 + R_2}$ for R .

Therefore, the time constant τ of the system is,

$$\frac{L(R_1 + R_2)}{R_1 \cdot R_2}$$

Step 5 of 5

Recall the unit step response of the system.

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-\frac{t}{\tau}}$$

Substitute $\frac{V}{R_1}$ for $i(\infty)$, 0 for $i(0)$ and $\frac{L(R_1 + R_2)}{R_1 \cdot R_2}$ for τ .

$$\begin{aligned} i(t) &= \frac{V}{R_1} + \left(0 - \frac{V}{R_1}\right)e^{-\frac{t}{\frac{L(R_1 + R_2)}{R_1 \cdot R_2}}} \\ &= \frac{V}{R_1} - \frac{V}{R_1}e^{-\frac{R_1 \cdot R_2 \cdot t}{L(R_1 + R_2)}} \\ &= \frac{V}{R_1} \left(1 - e^{-\frac{R_1 \cdot R_2 \cdot t}{L(R_1 + R_2)}}\right) \end{aligned}$$

Therefore, the step response of the circuit is,

$$\boxed{i(t) = \frac{V}{R_1} \left(1 - e^{-\frac{R_1 \cdot R_2 \cdot t}{L(R_1 + R_2)}}\right)}$$